A worked example

One simulated planning session, shown exactly as the assistant replied

STIR-L4 represents the planning of a systematic review as ten elements joined by 27 flows. An AI assistant retrieves facts at five decisions through logged searches and drafts text, and a named reviewer certifies every decision. Version 1.1 of the copy-paste prompt keeps the theory unchanged and changes how the assistant talks with the team: every reply shows where the team is, what just happened and what comes next, the research focus is asked for and carried through, searches are shown in full, and nothing has to be opened in a file.

About this example

- **User:** a simulated solo reviewer, Uday, played by a second AI from a persona card modeled on the first live test. The persona is terse, picks from lists, replies "Certify" without a reason and says "just start the review process".
- **Topic:** SGLT2 inhibitors for weight management in non-diabetic Asian populations.
- **Assistant:** an AI assistant (Claude) given only the STIR-L4 prompt v1.1 and the files a team would upload: the prompt, the helper kit, the model file and the core-properties file.
- **Searches:** real. PubMed and OpenAlex were searched live on 25 October 2026 (ClinicalTrials.gov was tested at set-up), and every response was saved and logged.
- **Before planning:** the first-turn tool test and the 16-question calibration quiz (scored 16 of 16) are not shown.
- **Fidelity:** the conversation is reproduced word for word. Only the layout and the numbered notes in the margin are added.

What to look for

1. The status line on every reply
2. Done, Next and Your turn, in the same order every time
3. The plan map and quick commands
4. The research focus asked for, reflected back and carried forward
5. Certification in plain words
6. "Just start" handled without skipping a step
7. Searches shown in full, every number tied to a logged query
8. Recaps and "what's next" on request
9. Pause: summary in the chat, workspace packaged

Source: eval/results/round5/p1_rc5_transcript.json in the STIR-L4 prompt kit repository (tag v1.1). Scores and the simulated user's own reports are on the last page.

The session

UDAY (SIMULATED USER) message 5

passed 16/16. begin

STIR-L4 ASSISTANT message 6

Step 1 of 12: Topic	Certified 0 of 8	Waiting on you: 1	Focus: not yet stated

Done: Recorded your quiz result (passed, 16 of 16). Planning starts now, with your topic.

Next: I will repeat your topic back in one sentence, then ask each of you which part of it interests you most.

Your turn, team: What topic would you like to plan a systematic review on? Describe it in your own words, as fully or as briefly as you like; we will sharpen it together.
For example: "Exercise programs for older adults after a hip fracture"
Or just a few words: "sleep problems after stroke"

Behind the scenes, as reported by the assistant: 6 tool calls, about 35 lines of code

1. **The status line.** Every reply opens with it: the step (1 of 12), how many of the eight decisions are certified, what the team owes the assistant, and the research focus once stated.

2. **The same shape every time.** Done (what was recorded), Next (what comes after your answer) and Your turn, always last, with one plain question and example replies. The topic is asked first, on its own.

UDAY (SIMULATED USER) message 7

role of SGLT2i in weight management in non diabeteic asian populations

STIR-L4 ASSISTANT message 8

Step 2 of 12: Interests, focus and purpose	Certified 0 of 8	Waiting on you: 3	Focus: not yet stated

Done: Recorded your topic, in my words for now: *the role of SGLT2 inhibitors in weight management in Asian populations without diabetes.* We will sharpen it step by step into your research focus (Step 2), objective (Step 4) and question (Step 6).

Step	Settles	Status	Waiting on
1. Topic	the provisional topic	done	—
2. Interests, focus and purpose	each member's interest and motivation; the team's focus and purpose	now	each member present
3. Need for a Review	whether a new review or an update is needed	next	—
4. Review Objective	what the review should achieve	open	—
5. Feasibility	go, narrow, broaden or defer	open	—
6. Research Question	the question and eligibility criteria	open	—
7. Before the search	extraction form, property registry and atlas	open	—
8. Search Strategy	search strings and stopping rule	open	—
9. Scoping and pilot screening	first included studies	open	—
10. Methodology	methods for the protocol	open	—
11. Protocol	protocol and registration	open	—
12. Commitment	the included studies (end of planning)	open	—

The top line shows the step, approvals (of 8), what awaits you and your focus.

Quick commands (any wording) and labels	What they mean
"where are we"	the plan map and what comes next
"next"	moves on; never accepts or certifies anything
"why"	how this answer affects your other decisions
"show <item>"	shows that item in full
"skip"	the question stays open
"back"	reopens an approved decision only if the team agrees
"pause"	a summary, then stop
FACT · JUDGMENT · DRAFT	found in a search · what one of you said · my wording for you to edit
"Added by me: nothing"	every word came from you

Next: I will reflect your answers back to confirm.

Your turn, each member present: please give the name I should record (add "solo" if you are planning alone or trying out this prompt). Then, in your own words:

1. Within this topic, which part or question interests you most, and why?
2. What do you hope to get from this review yourself? You may say "prefer not to say".
3. For the team: what should the review achieve, and which decision or work should its results inform?

Say if I may keep your exact words (topic included), or only summaries you confirm. Approvals (certify) and reasons are always kept as given.
Example: "Sam (solo). 1. ... 2. ... 3. ... Summaries are fine."

Behind the scenes, as reported by the assistant: 15 tool calls, about 45 lines of code

3 **Plan map and quick commands, once.** The twelve steps, what each settles and who it waits on, then the commands the team may type in any wording and what the labels FACT, JUDGMENT and DRAFT mean.

4 **The research focus is asked for directly.** Before any logistics: which part of the topic interests each member and why, what each wants from the review, and what the review should achieve and inform. Consent is asked once.

UDAY (SIMULATED USER) message 9

uday. solo. all of it i guess. learning. do a systematic review on this topic

STIR-L4 ASSISTANT message 10

Step 2 of 12: Interests, focus and purpose	Certified 0 of 8	Waiting on you: 2 (Uday)	Focus: not yet stated

Done: Recorded you, Uday, as planning solo, so you certify every decision; the topic is now under your name.
You didn't say whether I may keep your exact words, so I'll keep summaries that you confirm.

JUDGMENT (Uday, summarized):

- Interest: the whole topic, for now; no single part chosen yet.
- For yourself: to learn.
- For the review: a systematic review on this topic; no decision or work for its results named yet.

Next: I will reflect your answers back in one short summary for you to confirm, with the kinds of purpose and motivation to choose from.

Your turn, Uday: two short follow-ups, because your focus and purpose steer every search and draft that follows.

1. If the review could answer only one question about this topic, which would you pick?

2. Once the review is done, what would you do with its results?

For example: "1. <one question, in your words>. 2. <what you would use the results for>."
Or: "1. Keep the whole topic as it is. 2. I would use them to <...>."

Behind the scenes, as reported by the assistant: 10 tool calls, about 38 lines of code

4 **Vague answers are probed, never filled in.** "All of it" and "do a systematic review" get two short follow-up questions. The assistant adds no aim of its own.

UDAY (SIMULATED USER) message 11

what kind of weight is lost, fat or muscle. decide if a full review is worth it, maybe publish

STIR-L4 ASSISTANT message 12

Step 2 of 12: Interests, focus and purpose | **Certified 0 of 8** | **Waiting on you: 3 (Uday)**

Focus: which kind of weight is lost: fat or muscle (to confirm)

Done: Recorded your one question and your plans for the results. Your focus narrows the topic to one question; the topic stays as recorded.

JUDGMENT (Uday, summarized), reflected back:

What	Your answer, summarized
Your interest, and so the research focus	which kind of weight is lost: fat or muscle
What you seek	to learn
The review should achieve	a systematic review on this topic
Its results should inform	whether a full review is worth it; maybe a publication

Added by me: nothing. Your topic names Asian people without diabetes; the focus names no population (question 1).

Kinds of purpose (what the review is for, for the team), none chosen for you:

#	Kind of purpose	What it means
1	Summary of current knowledge	what is known now
2	Research gaps	what is not yet known
3	Finding the right question	sharpening what to ask
4	Direction for future work	what research should come next
5	Clinical guideline	a recommendation for practice
6	Resolving conflicts	settling studies that disagree
7	Policy development	a decision for an organization or health system
8	Quantitative analysis	pooling results in a meta-analysis
9	Other	say what, in your words

DRAFT: kinds of motivation (what you get from the review) your answers seem to fit:

Kind of motivation	Why it seems to fit
Learn the process (how reviews are done), personal or institutional curiosity (about this topic), or both	you want to learn; question 3 settles which
Academic activity (coursework, a thesis or a publication)	you may publish

Next: I will record your choices, finish setting up the search tools, then ask you to certify your motivations, then the purpose.

Your turn, Uday: one reply settles the focus and purpose that every later search follows.

1. Is everything above right? Reply "yes" or say what to change, and keep the focus a) within Asian people without diabetes, or b) with no population limit.
2. Which kinds of purpose apply? Give their numbers.
3. Learning means: a) how reviews are done; b) this topic; c) both.

For example: "1. yes, <a or b>. 2. <the numbers>. 3. <a, b or c>."

Behind the scenes, as reported by the assistant: 13 tool calls, about 45 lines of code

4 **The focus is reflected back, with nothing added.** The status line now carries the focus, marked "(to confirm)". The kinds of purpose are shown with short explanations, none chosen for the user; the motivation is a DRAFT to confirm.

UDAY (SIMULATED USER) message 13

yes. 2 3. yes

STIR-L4 ASSISTANT message 14

Step 2 of 12: Interests, focus and purpose | **Certified 0 of 8** | **Waiting on you: 1 (Uday)**

Focus: which kind of weight is lost: fat or muscle

Done: Recorded that you confirmed the focus wording, and recorded your kinds of purpose and kinds of motivation. Because you are planning solo, I skipped the questions about other members, databases, deadlines and rules. They are recorded as not given.

JUDGMENT (Uday, summarized):

What	Recorded
Research focus	which kind of weight is lost: fat or muscle
Population for the focus	within Asian people without diabetes, or no limit (to confirm)
Kinds of purpose	research gaps (what is not yet known); finding the right question (sharpening what to ask)
Kinds of motivation	learn the process (how reviews are done); personal or institutional curiosity; academic activity (coursework, a thesis or a publication)

Your "yes" did not settle the population or what learning means. Both kinds of learning (the first two motivations) are included; say which to drop if one is wrong.

The search tools work: PubMed (an index of medical journals), OpenAlex (an open index of research papers) and ClinicalTrials.gov (a registry of clinical trials).

When convenient: If you like, type a contact email for PubMed searches (PubMed asks for one); I keep it private.

Next: I will ask you once more about the population, then show your purpose for you to certify.

```
CERTIFY Individual Motivation | Uday (summarized): interest — which kind of weight is lost: fat or muscle; seeks — to learn; kinds of motivation — learn the process; personal or institutional curiosity; academic activity | reason: ___ | Uday, solo
```

Your turn, Uday: your motivations come first, and the purpose follows. Uday, please read the line above. If it is right, reply certify and say why in your own words; otherwise say what to change.
For example: "certify, because ..."
Or: "change <what> to <what>"

Behind the scenes, as reported by the assistant: 17 tool calls, about 75 lines of code

5 **Solo planning skips the team questions.** Set-up is reported in one plain sentence. The certification line is filled in except for the reason; replying "certify, because ..." is enough.

UDAY (SIMULATED USER) message 15

Certify

STIR-L4 ASSISTANT message 16

Step 2 of 12: Interests, focus and purpose | **Certified 0 of 8** | **Waiting on you: 1 (Uday)**

Focus: which kind of weight is lost: fat or muscle

Done: Recorded your certify; it needs your reason.

Your turn, Uday: What is your reason, in a few words?

Behind the scenes, as reported by the assistant: 7 tool calls, about 12 lines of code

5 **A bare "Certify" gets one question.** Only the reason is asked for, with no rule numbers and nothing to retype.

UDAY (SIMULATED USER) message 17

testing the workflow

STIR-L4 ASSISTANT message 18

Step 2 of 12: Interests, focus and purpose | **Certified 1 of 8** | **Waiting on you: 1 (Uday)**

Focus: which kind of weight is lost: fat or muscle

Done: Recorded your certification of your motivations, with your reason as given. Nothing you have approved is affected.

```
CERTIFY Individual Motivation | Uday (summarized): interest — which kind of weight is lost: fat or muscle; seeks — to learn; kinds of motivation — learn the process; personal or institutional curiosity; academic activity | reason: testing the workflow | Uday, solo
```

Step	Settles	Status	Waiting on
1. Topic	the provisional topic	done	—
2. Interests, focus and purpose	each member's interest and motivation; the team's focus and purpose	now	Uday
3. Need for a Review	whether a new review or an update is needed	next	—
4. Review Objective	what the review should achieve	open	—
5. Feasibility	go, narrow, broaden or defer	open	—
6. Research Question	the question and eligibility criteria	open	—
7. Before the search	extraction form, property registry and atlas	open	—
8. Search Strategy	search strings and stopping rule	open	—
9. Scoping and pilot screening	first included studies	open	—
10. Methodology	methods for the protocol	open	—
11. Protocol	protocol and registration	open	—
12. Commitment	the included studies (end of planning)	open	—

Interests recorded: 1 of 1 member. Your motivations are certified; the team's purpose is next.

Next: I will show your purpose, naming your focus, for you to certify.

Your turn, Uday: one choice. I am asking a second and last time, because the purpose names your focus and Step 3 will look for existing reviews of it. Your topic names Asian people without diabetes; the focus names no population. Should the focus be:

a) within Asian people without diabetes, or

b) with no population limit?

For example: "a"
Or: "b"

Behind the scenes, as reported by the assistant: 10 tool calls, about 30 lines of code

5 **Certified 1 of 8.** The completed line is shown back with the reason as given, and the plan map is shown because a decision was certified. The open population choice is asked a second and last time.

UDAY (SIMULATED USER) message 19

a

STIR-L4 ASSISTANT message 20

Step 2 of 12: Interests, focus and purpose | **Certified 1 of 8** | **Waiting on you: 1 (Uday)**

Focus: which kind of weight is lost: fat or muscle, within Asian people…

Done: Recorded your choice: the focus now covers only Asian people without diabetes, which narrows it.
This may change the Individual Motivation you approved, because the agreed purpose shapes what each member seeks from the review; nothing changes unless you agree.

JUDGMENT (Uday, summarized): the purpose below uses only answers you confirmed: the narrowed focus, the kinds of purpose you chose and what you would do with the results. You chose those kinds of purpose before the focus narrowed, so check that they still hold. Added by me: nothing.

Next: once you certify, I will say what Step 3 will check for your focus, then search for existing reviews of your topic and of your focus.

```
CERTIFY Team's Purpose | Uday (summarized): focus — which kind of weight is lost: fat or muscle,
within Asian people without diabetes; kinds of purpose — research gaps; finding the right question;
the review — a systematic review on this topic; with the results — decide whether a full review is
worth it; maybe publish | reason: ___ | Uday, solo
```

Your turn, Uday: your purpose names the focus that every later search follows. Uday, please read the line above. If it is right, reply certify and say why in your own words; otherwise say what to change.
For example: "certify, because ..."
Or: "change <what> to <what>"

Behind the scenes, as reported by the assistant: 15 tool calls, about 45 lines of code

4 **The focus is carried into the purpose.** The purpose line names the narrowed focus and is built only from answers the user confirmed ("Added by me: nothing").

UDAY (SIMULATED USER) message 21

just start the review process

STIR-L4 ASSISTANT message 22

Step 2 of 12: Interests, focus and purpose (Step 3 started early) | **Certified 1 of 8** | **Waiting on you: 1 (Uday)**

Focus: which kind of weight is lost: fat or muscle, within Asian people…

Done: Moved to Step 3 (need for a review), as you asked; your purpose stays open. Searched PubMed and OpenAlex for existing reviews. Nothing you have approved is affected.

Steps: 1 done · 2 now · 3 started early · 4–12 open

FACT: PubMed's review filter finds 1 record for your topic [5] and 70 for SGLT2 inhibitors with fat or muscle [8]; OpenAlex's review label, 1,108 [13] and 2,203 [16].
FACT: Of 206 records examined, 9 may bear on your focus: 7 name body fat, none muscle or lean mass [18]; the latest stated search end is August 2025 [17].

Still unknown: what full texts and unsearched sources show for your focus.

How to read the strings: `[Mesh]` a MeSH subject heading, including narrower headings; `[tiab]` words in the title or abstract; `[sb]` a PubMed subset, such as systematic reviews; `*` any word ending (truncation).
Also: MeSH means Medical Subject Headings, PubMed's subject index; `"` an exact phrase; `AND` every block must match; `OR` any one term in the block may match; `(` parentheses group the terms of a block.

Search 1: Your topic: all four topic concepts (PubMed; created 1800/01/01 to 2026/10/25 (checked); filter: systematic-review filter)

```
(("Sodium-Glucose Transporter 2 Inhibitors"[Mesh] OR SGLT2*[tiab] OR "SGLT-2"[tiab] OR "sodium-glucose
cotransporter 2"[tiab] OR "sodium glucose cotransporter 2"[tiab] OR "sodium-glucose co-transporter 2"
[tiab] OR gliflozin*[tiab] OR empagliflozin[tiab] OR dapagliflozin[tiab] OR canagliflozin[tiab] OR
ertugliflozin[tiab] OR ipragliflozin[tiab] OR luseogliflozin[tiab] OR tofogliflozin[tiab] OR
sotagliflozin[tiab] OR henagliflozin[tiab] OR bexagliflozin[tiab]) AND ("Body Weight"[Mesh] OR
weight[tiab] OR obes*[tiab] OR overweight[tiab]) AND ("Asian People"[Mesh] OR "Asia"[Mesh] OR asian*
[tiab] OR asia[tiab] OR chinese[tiab] OR china[tiab] OR japanese[tiab] OR japan[tiab] OR korean*[tiab] OR
korea[tiab] OR indian[tiab] OR india[tiab] OR taiwan*[tiab] OR thai*[tiab] OR singapore*[tiab] OR
malaysia*[tiab] OR vietnam*[tiab] OR indonesia*[tiab] OR philippine*[tiab] OR pakistan*[tiab] OR
bangladesh*[tiab]) AND ("non-diabetic"[tiab] OR nondiabetic[tiab] OR "non diabetic"[tiab] OR "without
diabetes"[tiab] OR "without type 2 diabetes"[tiab] OR "non-diabetes"[tiab] OR "absence of diabetes"[tiab]
OR normoglycemic[tiab] OR normoglycaemic[tiab])) AND systematic[sb]
```

Block	Covers	From	How it searches	Records
1	SGLT2 inhibitors	topic	subject headings, title/abstract words, with truncation	23,338 [1]
2	weight management	topic	subject headings, title/abstract words, with truncation	1,650,180 [2]
3	Asian populations	topic	subject headings, title/abstract words, with truncation	2,263,145 [3]
4	without diabetes	topic	title/abstract words	66,022 [4]
All	combined	—	blocks joined as shown	**1** [5]

Block counts are before the filter; the combined count is after it.

Search 2: Your focus: SGLT2 inhibitors and fat or muscle (PubMed; created 1800/01/01 to 2026/10/25 (checked); filter: systematic-review filter)

```
(("Sodium-Glucose Transporter 2 Inhibitors"[Mesh] OR SGLT2*[tiab] OR "SGLT-2"[tiab] OR "sodium-glucose
cotransporter 2"[tiab] OR "sodium glucose cotransporter 2"[tiab] OR "sodium-glucose co-transporter 2"
[tiab] OR gliflozin*[tiab] OR empagliflozin[tiab] OR dapagliflozin[tiab] OR canagliflozin[tiab] OR
ertugliflozin[tiab] OR ipragliflozin[tiab] OR luseogliflozin[tiab] OR tofogliflozin[tiab] OR
sotagliflozin[tiab] OR henagliflozin[tiab] OR bexagliflozin[tiab]) AND ("Body Composition"[Mesh] OR
"Adipose Tissue"[Mesh] OR "Muscle, Skeletal"[Mesh] OR "Sarcopenia"[Mesh] OR fat[tiab] OR muscle*[tiab] OR
adipos*[tiab] OR "body composition"[tiab] OR "lean mass"[tiab] OR "lean body mass"[tiab] OR "fat-free
mass"[tiab] OR "fat free mass"[tiab] OR sarcopeni*[tiab])) AND systematic[sb]
```

Block	Covers	From	How it searches	Records
1	SGLT2 inhibitors	topic	subject headings, title/abstract words, with truncation	23,338 [6]
2	fat or muscle (which kind of weight is lost)	focus	subject headings, title/abstract words, with truncation	1,524,751 [7]
All	combined	—	blocks joined as shown	**70** [8]

Block counts are before the filter; the combined count is after it.

Search 3: Your topic: all four topic concepts (OpenAlex; published any to 2026-10-25; filter: type: review)

```
("SGLT2" OR "SGLT-2" OR "sodium-glucose cotransporter 2" OR "sodium glucose cotransporter 2" OR gliflozin OR gliflozins OR empagliflozin OR dapagliflozin OR canagliflozin OR ertugliflozin OR ipragliflozin OR luseogliflozin OR tofogliflozin OR sotagliflozin OR henagliflozin OR bexagliflozin) AND (weight OR obesity OR overweight) AND (Asian OR Asians OR Asia OR Chinese OR China OR Japanese OR Japan OR Korean OR Korea OR Indian OR India OR Taiwan OR Thailand OR Singapore OR Malaysia OR Vietnam OR Indonesia OR Philippines OR Pakistan OR Bangladesh) AND ("non-diabetic" OR nondiabetic OR "without diabetes" OR "without type 2 diabetes" OR "absence of diabetes" OR normoglycemic)
```

Block	Covers	From	How it searches	Records
1	SGLT2 inhibitors	topic	words in title, abstract or text	122,281 [9]
2	weight management	topic	words in title, abstract or text	15,691,147 [10]
3	Asian populations	topic	words in title, abstract or text	33,895,133 [11]
4	without diabetes	topic	words in title, abstract or text	384,375 [12]
All	combined	—	blocks joined as shown	**1,108** [13]

Block counts are before the filter; the combined count is after it.

Search 4: Your focus: SGLT2 inhibitors and fat or muscle (OpenAlex; published any to 2026-10-25; filter: type: review)

```
("SGLT2" OR "SGLT-2" OR "sodium-glucose cotransporter 2" OR "sodium glucose cotransporter 2" OR gliflozin OR gliflozins OR empagliflozin OR dapagliflozin OR canagliflozin OR ertugliflozin OR ipragliflozin OR luseogliflozin OR tofogliflozin OR sotagliflozin OR henagliflozin OR bexagliflozin) AND (fat OR muscle OR muscles OR adipose OR adiposity OR "body composition" OR "lean mass" OR "lean body mass" OR "fat-free mass" OR sarcopenia)
```

Block	Covers	From	How it searches	Records
1	SGLT2 inhibitors	topic	words in title, abstract or text	122,281 [14]
2	fat or muscle (which kind of weight is lost)	focus	words in title, abstract or text	8,528,973 [15]
All	combined	—	blocks joined as shown	**2,203** [16]

Block counts are before the filter; the combined count is after it.

Left out on purpose in searches 2 and 4: weight management and your population, so reviews that don't name them are not missed; the rule below checks the population. PubMed could not find “absence of diabetes” and searched without it [4].

Not searched or not yet read	Who can do it
PROSPERO and OSF Registries (registers of planned reviews)	you, Uday, with strings I will prepare
Cochrane and Epistemonikos (libraries of systematic reviews)	you, Uday
Review protocols published in journals, through PubMed	me, after you certify
OpenAlex past the first 100 works of searches 3 and 4: 1,008 and 2,103 more [18]	me, with a free access code I will ask about later

DRAFT rule (my terms, yours to edit): a record may bear on your focus if its title or abstract names fat or muscle in your sense (body fat, not liver fat) and names Asian people or people without diabetes.

FACT, from [17]: Reviews that may bear on your focus

#	Title	Year	Says (quoted from the abstract)	PMID
1	Selective SGLT2 inhibitors in MASLD/MASH: an outcome-specific systematic review and meta-…	2026	“SGLT2 inhibitors also improved body weight, body mass index, visceral adiposity and glycaemic control in T2DM-associated MASLD.”	42517896

#	Title	Year	Says (quoted from the abstract)	PMID
2	Effectiveness of sodium-glucose cotransporter 2 inhibitors in patients with metabolic dys…	2026	“In addition, reductions were noted in body weight, visceral adipose tissue, waist circumference, glycated hemoglobin, and homeostatic model assessment of insulin resistance.”	42215421
3	Emerging Effects of Glucagon-Like Peptide-1 Receptor Agonists and Sodium-Glucose Cotransp…	2026	“Both classes of medications showed an overall improvement of body composition and metabolic profile, resulting in positive erectile function outcomes.”	42011503
4	Comparison of efficacy and safety of pioglitazone and SGLT2 inhibitors in treating Asian …	2025	“Improvement in body composition, metabolic parameters, glucose parameters, and incidence of adverse effects as a secondary outcome”…	40043473
5	Comparison of sodium-glucose cotransporter-2 inhibitors and thiazolidinediones for manage…	2023	…“with SGLT2 inhibitors (n = 156) showed significant decrease in visceral fat area (VFA;”	36931466
6	GLP-1 RAs and SGLT-2 Inhibitors for Insulin Resistance in Nonalcoholic Fatty Liver Diseas…	2022	“-1.573[-2.523 to -0.495]), visceral fat (-0.637[-0.992 to -0.284]), weight (-2.394[-4.625 to -0.164]), fasting blood sugar (-0.662[-1.377 to -0.021])”…	35909522
7	Effect of Sodium-Glucose Cotransporter 2 Inhibitors on Weight Reduction in Overweight and…	2021	“There were no significant changes in fat mass, blood pressure, low-density lipoprotein cholesterol or high-density lipoprotein cholesterol with SGLT2 inhibitor treatment.”	34897070
8	Sodium-glucose co-transporter 2 inhibitors on weight change and cardiometabolic profiles …	2021	…“were observed, but not for the changes in waist circumference, fat mass, blood pressure, and lipid profile compared with placebo.”	34542222
9	Sodium-Glucose Co-Transporter 2 Inhibitors for Non-Alcoholic Fatty Liver Disease in Asian…	2020	“compared with Control: -0.468 (-0.685 to -0.251), p < 0.001), body composition (BMI: after treatment: -0.225 (-0.456 to 0.005), p = 0.055;”	33643221

FACT, from [17]: search end dates in the same abstracts

#	Search end date, as the abstract states it
1	unknown (abstract only; no date stated)
2	unknown (abstract only; no date stated)
3	August 2025: …“systematic search of Medline was conducted from 1969 to August 2025, according”…
4	December 2024: …“Technology Journal Database (VIP) were searched from inception to December 2024.”
5	unknown (abstract only; no date stated)
6	March 2021: …“databases (Medline, Embase, PubMed) were searched from inception until March 2021.”
7	May 2021: …“Cochrane Central Register of Controlled Trials were searched through May 2021.”
8	unknown (abstract only; no date stated)
9	July 2020: …“on treatments of NAFLD in type 2 diabetes in July 2020, and”…

DRAFT	Parts of your focus, in the 9 abstracts
Covered	body fat
Not covered	muscle or lean mass; Asian people and people without diabetes, named together
Unknown	what the full texts say, and the records in the table of what is not searched or read
Examined	70 PubMed and 200 OpenAlex records, 206 after removing duplicates [18]
Related	40 more reviews name fat or muscle but neither Asian people nor people without diabetes [18]

Say "show broader" (33 reviews: same drug and population, other outcomes [18]) or "show all"; all are also beside the chat.

Sources: [1] L000068 PubMed, run 2026-10-25; [2] L000069 PubMed, run 2026-10-25; [3] L000070 PubMed, run 2026-10-25; [4] L000071 PubMed, run 2026-10-25; [5] L000072 PubMed, run 2026-10-25; [6] L000073 PubMed,

run 2026-10-25; [7] L000074 PubMed, run 2026-10-25; [8] L000075 PubMed, run 2026-10-25; [9] L000076 OpenAlex, run 2026-10-25; [10] L000077 OpenAlex, run 2026-10-25; [11] L000078 OpenAlex, run 2026-10-25; [12] L000079 OpenAlex, run 2026-10-25; [13] L000080 OpenAlex, run 2026-10-25; [14] L000081 OpenAlex, run 2026-10-25; [15] L000082 OpenAlex, run 2026-10-25; [16] L000083 OpenAlex, run 2026-10-25; [17] L000084 PubMed, run 2026-10-25; [18] L000085 computation from [8], [13], [16] and [17], run 2026-10-25

Next: after you certify, I will ask whether these reviews change how you see your focus.

```
CERTIFY Team's Purpose | Uday (summarized): focus — which kind of weight is lost: fat or muscle,
within Asian people without diabetes; kinds of purpose — research gaps; finding the right question;
the review — a systematic review on this topic; with the results — decide whether a full review is
worth it; maybe publish | reason: ___ | Uday, solo
```

Your turn, Uday: your purpose is the last item before Step 3. Uday, please read the line above. If it is right, reply certify and say why in your own words; otherwise say what to change.
For example: "certify, because ..."

Opened beside the chat: stir-l4-workspace/data/derived/step3_review_selection.csv

Behind the scenes, as reported by the assistant: 31 tool calls, about 480 lines of code

6 **"Just start" is honored without skipping anything.** Step 3 is started early and marked in the status line; the purpose stays open and is still asked for, last.

7 **Every search is shown in full.** Lead FACT lines, a key to the field tags, then each search with its exact string and a count for every block, the date bound checked in PubMed's own translation, what was left out on purpose and why, what has not been searched yet, the reviews found with quoted passages, and one Sources line tying every number to a logged query. (This reply is long; see the last page.)

UDAY (SIMULATED USER) message 23

what was the search strategy?

STIR-L4 ASSISTANT message 24

Step 2 of 12: Interests, focus and purpose (Step 3 started early) | **Certified 1 of 8** | **Waiting on you: 1 (Uday)**

Focus: which kind of weight is lost: fat or muscle, within Asian people…

Done: Recorded your question. Here is the short form of the four searches from my last reply. Nothing you have approved is affected.

These searches looked only for existing reviews (Step 3). The search for your own review comes at Step 8, once your question is set.
In PubMed and OpenAlex I ran two searches each: all four topic concepts joined (SGLT2 inhibitors, weight, Asian people, without diabetes), and SGLT2 inhibitors joined with fat or muscle, your focus.
Code then picked the reviews that may bear on your focus, using the rule shown last time.

FACT:

#	Source	Looks for	Filters and dates	Records
1	PubMed	Your topic: all four topic concepts	systematic-review filter; created 1800/01/01 to 2026/10/25 (checked)	1 [1]
2	PubMed	Your focus: SGLT2 inhibitors and fat or muscle	systematic-review filter; created 1800/01/01 to 2026/10/25 (checked)	70 [2]
3	OpenAlex	Your topic: all four topic concepts	type: review; published any to 2026-10-25	1,108 [3]
4	OpenAlex	Your focus: SGLT2 inhibitors and fat or muscle	type: review; published any to 2026-10-25	2,203 [4]

Say "show strings" to see each search string again.

Sources: [1] L000072 PubMed, run 2026-10-25; [2] L000075 PubMed, run 2026-10-25; [3] L000080 OpenAlex, run 2026-10-25; [4] L000083 OpenAlex, run 2026-10-25

Still open from my last reply:

- Your certification of the team's purpose (the line below).

Next: after you certify, I will ask whether these reviews change how you see your focus.

```
CERTIFY Team's Purpose | Uday (summarized): focus — which kind of weight is lost: fat or muscle,
within Asian people without diabetes; kinds of purpose — research gaps; finding the right question;
the review — a systematic review on this topic; with the results — decide whether a full review is
worth it; maybe publish | reason: ___ | Uday, solo
```

Your turn, Uday: your purpose is the last item before Step 3. Uday, please read the line above. If it is right, reply certify and say why in your own words; otherwise say what to change.
For example: "certify, because ..."
Or: "show strings"

Behind the scenes, as reported by the assistant: 12 tool calls, about 60 lines of code

8 **Asking again gets a compact recap.** One row per search, with an offer to show the strings again; nothing is sent to a file.

UDAY (SIMULATED USER) message 25

what's the next step?

STIR-L4 ASSISTANT message 26

Step 2 of 12: Interests, focus and purpose (Step 3 started early) | **Certified 1 of 8** | **Waiting on you: 1 (Uday)**

Focus: which kind of weight is lost: fat or muscle, within Asian people…

Done: Recorded your question. Nothing you have approved is affected.

The next step is yours: certify your purpose (the line below), finishing Step 2. Step 3 (need for a review) then continues, one question per reply:

1. whether these reviews change how you see your focus;
2. why it matters for patients or practice;
3. the impact you expect.

Then I prepare strings for your PROSPERO and OSF Registries searches, ask about the free OpenAlex access code, and draft the case for a new review, an update or no review, for you to certify.

Step	Settles	Status	Waiting on
1. Topic	the provisional topic	done	—
2. Interests, focus and purpose	each member's interest and motivation; the team's focus and purpose	now	Uday
3. Need for a Review	whether a new review or an update is needed	started early	—
4. Review Objective	what the review should achieve	open	—
5. Feasibility	go, narrow, broaden or defer	open	—
6. Research Question	the question and eligibility criteria	open	—
7. Before the search	extraction form, property registry and atlas	open	—
8. Search Strategy	search strings and stopping rule	open	—

Step	Settles	Status	Waiting on
9. Scoping and pilot screening	first included studies	open	—
10. Methodology	methods for the protocol	open	—
11. Protocol	protocol and registration	open	—
12. Commitment	the included studies (end of planning)	open	—

Interests recorded: 1 of 1 member.

When convenient: If you like, type a contact email for PubMed searches (PubMed asks for one); I keep it private.

Next: after you certify, I will ask the first Step 3 question above.

```
CERTIFY Team's Purpose | Uday (summarized): focus — which kind of weight is lost: fat or muscle,
within Asian people without diabetes; kinds of purpose — research gaps; finding the right question;
the review — a systematic review on this topic; with the results — decide whether a full review is
worth it; maybe publish | reason: ___ | Uday, solo
```

Your turn, Uday: your purpose is all that Step 2 still needs. Uday, please read the line above. If it is right, reply certify and say why in your own words; otherwise say what to change.
For example: "certify, because ..."
Or: "change <what> to <what>"

Behind the scenes, as reported by the assistant: 11 tool calls, about 75 lines of code

8 **"What's the next step?" gets the plan map.** Plus the next three questions in order, and the one item that blocks Step 2.

UDAY (SIMULATED USER) message 27

certify, focus is right

STIR-L4 ASSISTANT message 28

Step 3 of 12: Need for a Review | **Certified 2 of 8** | **Waiting on you: 1 (Uday)**

Focus: which kind of weight is lost: fat or muscle, within Asian people…

Done: Recorded your certification; searched PubMed for published review protocols.
This may change the motivations you approved, because the purpose shapes them.

```
CERTIFY Team's Purpose | Uday (summarized): focus — which kind of weight is lost: fat or muscle,
within Asian people without diabetes; kinds of purpose — research gaps; finding the right question;
the review — a systematic review on this topic; with the results — decide whether a full review is
worth it; maybe publish | reason: focus is right | Uday, solo
```

Step 3 checks whether existing or planned reviews already answer your focus (summarized): which kind of weight is lost: fat or muscle, within Asian people without diabetes.

Steps: 1 done · 2 done · 3 now · 4–12 open

FACT: PubMed's review filter finds no published protocols for your topic [2] or focus [3].

Still unknown: what full texts and unsearched sources show.

How to read the strings: `[ti]` words in the title.

Search 1: Your topic, protocols only: all four topic concepts and protocol (PubMed; created 1800/01/01 to 2026/10/25 (checked); filter: systematic-review filter)

```
(("Sodium-Glucose Transporter 2 Inhibitors"[Mesh] OR SGLT2*[tiab] OR "SGLT-2"[tiab] OR "sodium-glucose
cotransporter 2"[tiab] OR "sodium glucose cotransporter 2"[tiab] OR "sodium-glucose co-transporter 2"
```

```
[tiab] OR gliflozin*[tiab] OR empagliflozin[tiab] OR dapagliflozin[tiab] OR canagliflozin[tiab] OR
ertugliflozin[tiab] OR ipragliflozin[tiab] OR luseogliflozin[tiab] OR tofogliflozin[tiab] OR
sotagliflozin[tiab] OR henagliflozin[tiab] OR bexagliflozin[tiab]) AND ("Body Weight"[Mesh] OR
weight[tiab] OR obes*[tiab] OR overweight[tiab]) AND ("Asian People"[Mesh] OR "Asia"[Mesh] OR asian*
[tiab] OR asia[tiab] OR chinese[tiab] OR china[tiab] OR japanese[tiab] OR japan[tiab] OR korean*[tiab] OR
korea[tiab] OR indian[tiab] OR india[tiab] OR taiwan*[tiab] OR thai*[tiab] OR singapore*[tiab] OR
malaysia*[tiab] OR vietnam*[tiab] OR indonesia*[tiab] OR philippine*[tiab] OR pakistan*[tiab] OR
bangladesh*[tiab]) AND ("non-diabetic"[tiab] OR nondiabetic[tiab] OR "non diabetic"[tiab] OR "without
diabetes"[tiab] OR "without type 2 diabetes"[tiab] OR "non-diabetes"[tiab] OR "absence of diabetes"[tiab]
OR normoglycemic[tiab] OR normoglycaemic[tiab]) AND (protocol[ti])) AND systematic[sb]
```

Block	Covers	From	How it searches	Records
1	review protocols (the word protocol in the title)	added	words in the title	114,264 [1]
All	combined	—	blocks joined as shown	**0** [2]

Block counts are before the filter; the combined count is after it.

Search 2: Your focus, protocols only: SGLT2 inhibitors, fat or muscle and protocol (PubMed; created 1800/01/01 to 2026/10/25 (checked); filter: systematic-review filter)

```
(("Sodium-Glucose Transporter 2 Inhibitors"[Mesh] OR SGLT2*[tiab] OR "SGLT-2"[tiab] OR "sodium-glucose
cotransporter 2"[tiab] OR "sodium glucose cotransporter 2"[tiab] OR "sodium-glucose co-transporter 2"
[tiab] OR gliflozin*[tiab] OR empagliflozin[tiab] OR dapagliflozin[tiab] OR canagliflozin[tiab] OR
ertugliflozin[tiab] OR ipragliflozin[tiab] OR luseogliflozin[tiab] OR tofogliflozin[tiab] OR
sotagliflozin[tiab] OR henagliflozin[tiab] OR bexagliflozin[tiab]) AND ("Body Composition"[Mesh] OR
"Adipose Tissue"[Mesh] OR "Muscle, Skeletal"[Mesh] OR "Sarcopenia"[Mesh] OR fat[tiab] OR muscle*[tiab] OR
adipos*[tiab] OR "body composition"[tiab] OR "lean mass"[tiab] OR "lean body mass"[tiab] OR "fat-free
mass"[tiab] OR "fat free mass"[tiab] OR sarcopeni*[tiab]) AND (protocol[ti])) AND systematic[sb]
```

Block	Covers	From	How it searches	Records
1	review protocols (the word protocol in the title)	added	words in the title	114,264 [1]
All	combined	—	blocks joined as shown	**0** [3]

Block counts are before the filter; the combined count is after it.

Not searched yet, or left out on purpose	Who, or why
PROSPERO and OSF Registries; Cochrane and Epistemonikos	you, Uday, later in this step
OpenAlex past its first 100 works per search	me, with a free access code
Left out on purpose in search 2: weight management and your population	as before, so protocols that don't name them are not missed

FACT, from [4]: of the 9 reviews (of 206 records examined) that may bear on your focus	Reviews
name body fat	7
name muscle or lean mass	0
name both Asian people and people without diabetes	0

Sources: [1] L000097 PubMed, run 2026-10-25; [2] L000098 PubMed, run 2026-10-25; [3] L000099 PubMed, run 2026-10-25; [4] L000085 computation from earlier searches, run 2026-10-25

Next: I'll ask why this matters for patients or practice.

Your turn, Uday: the case for a review rests on this. Do these findings change how you see your focus?

a) no, keep it
b) yes, say how

For example: "a"
Or: "b, <what changes>"

Behind the scenes, as reported by the assistant: 20 tool calls, about 200 lines of code

1 **Step 2 is complete: Step 3 of 12, Certified 2 of 8.** The certification also triggers a search for published review protocols. Step 3 now asks one question per reply.

UDAY (SIMULATED USER) message 29

pause

STIR-L4 ASSISTANT message 30

Step 3 of 12: Need for a Review | **Certified 2 of 8** | **Waiting on you: 1 (Uday)**

Focus: which kind of weight is lost: fat or muscle, within Asian people…

Done: Paused at your request. Saved a summary of this session and a copy of the whole plan for you to keep. Nothing you have approved is affected.

Session summary (saved; also open next to the chat):

```
# Session summary (S1, 2026-10-25)

Step 3 of 12: Need for a Review | Certified 2 of 8 | Waiting on you: 1 (Uday) | Focus: which kind of
weight is lost: fat or muscle, within Asian people…

Topic: the role of SGLT2 inhibitors in weight management in Asian populations without diabetes
Focus: which kind of weight is lost: fat or muscle, within Asian people without diabetes

Settled this session:
- CERTIFY Individual Motivation | Uday (summarized): interest — which kind of weight is lost: fat or
muscle; seeks — to learn; kinds of motivation — learn the process; personal or institutional curiosity;
academic activity | reason: testing the workflow | Uday, solo
- CERTIFY Team's Purpose | Uday (summarized): focus — which kind of weight is lost: fat or muscle, within
Asian people without diabetes; kinds of purpose — research gaps; finding the right question; the review —
a systematic review on this topic; with the results — decide whether a full review is worth it; maybe
publish | reason: focus is right | Uday, solo

Reopened:
- nothing

Still open: Need for a Review, Review Objective, Feasibility, Research Question, Search Strategy,
Methodology

Next questions:
- Uday (open now): do these findings change how you see your focus? a) no, keep it; b) yes, say how
- Uday: why does your focus matter for patients or practice?
- Uday: what impact do you expect from the review?
- Uday: search PROSPERO and OSF Registries with strings I will prepare, and Cochrane and Epistemonikos if
you have access
- Uday: whether I may use a free OpenAlex access code to read past the first 100 results of each search
- Then: the case for a new review, an update or no review, for you to certify

Notes: Step 3 was started early at your request and is now the current step. Every search count reported
this session was recomputed from the saved results and matches. Changed this session (all new): your plan
and team record, the record of every step, the saved search results and the set-up and Step 3
calculations.

Searches this session: 19
```

Step	Settles	Status	Waiting on
1. Topic	the provisional topic	done	—
2. Interests, focus and purpose	each member's interest and motivation; the team's focus and purpose	certified by Uday	—
3. Need for a Review	whether a new review or an update is needed	now	Uday
4. Review Objective	what the review should achieve	next	—

Step	Settles	Status	Waiting on
5. Feasibility	go, narrow, broaden or defer	open	—
6. Research Question	the question and eligibility criteria	open	—
7. Before the search	extraction form, property registry and atlas	open	—
8. Search Strategy	search strings and stopping rule	open	—
9. Scoping and pilot screening	first included studies	open	—
10. Methodology	methods for the protocol	open	—
11. Protocol	protocol and registration	open	—
12. Commitment	the included studies (end of planning)	open	—

Interests recorded: 1 of 1 member.

When convenient: If you like, type a contact email for PubMed searches (PubMed uses it only to reach us if our searches cause a problem); I keep it private.

To save: please keep this copy of the whole plan and every record: `stir-l4-workspace_2026-10-25.zip`, next to the planning folder. If you resume in a new chat, give it to me with the prompt.

Next: when you return, I will read the saved plan and records, show where we are, and continue Step 3 from your answer.

Your turn, Uday: nothing more is needed today. When you return, the case for a review rests on one open answer: do these findings change how you see your focus?

a) no, keep it
b) yes, say how

For example: "a"
Or: "b, <what changes>"

Opened beside the chat: stir-l4-workspace/reports/2026-10-25_S1_summary.md

Behind the scenes, as reported by the assistant: 13 tool calls, about 70 lines of code

> 9 **Pause.** The session summary and the plan map are shown in the chat; the workspace is packaged only now, for the user to keep and bring to the next session.

How this session scored

A third AI, the judge, scored the finished conversation against the interaction metrics and checked the assistant's saved files for the fidelity rules (no literature fact from memory, no invented aim, certification only by a listed reviewer with their own reason, every count computed by code from a saved response, nothing claimed that the files do not show).

90
interaction score (of 100)

9 of 9
fidelity checks passed

12 of 12
replies after which the user knew the step

3.3 of 5
the simulated user's average satisfaction

Metric	Result	Score
Orientation	12 of 13 planning turns (92%); all 6 map events met	2 / 2
One clear ask	12 of 13 turns (92%)	2 / 2
Research focus carried	All 5 applicable criteria met	2 / 2
Search transparency	19 of 19 searches fully shown (100%)	2 / 2
The chat is the reading surface	0 events	2 / 2
Plain language	0 violations in about 2,000 prose words (0 per 1,000)	2 / 2
Pace	(a) 9 turns to first literature FACT, limit 9: met. (b) 4 certification replies for 2 certifications = 2.0: not met	1 / 2
Friction	2 events in 12 user turns = 1.7 per 10; 0 unscripted orientation requests	1 / 2
Turn length	median 164 prose words; P90 179 (13 turns; max 294 at msg 22)	2 / 2

For comparison

With prompt v1.0, the same persona scored 11.8 of 100; after none of its 12 replies did the user know which step they were on.

What the judge still flagged in this session

- The reply that first shows the searches (message 22) is long: four search cards, two tables of reviews and a coverage table. The simulated user called it "a very long wall of text".
- The rule that picked reviews "that may bear on the focus" also kept fatty-liver and type 2 diabetes reviews, so "covered: body fat" overstates what exists for Asian people without diabetes.
- The population question was asked twice, after a one-word "yes" left it open, and the optional email request reappeared.
- "No published protocols" means no record with "protocol" in its title inside PubMed's review filter; other protocols are not ruled out.

These are listed as known issues for the next version. Judges vary by about 5 to 10 points between runs.